PROJECT REPORT

F1 Race Intelligence Dashboard

Interactive Formula 1 Analytics using Tableau and Flask

Detail	Information
Student Name	Vijeta Nagda
Team ID	EN25CS3011046
Project Title	F1 Race Intelligence Dashboard
Tools Used	Tableau Public, Python Flask, Render
Project Duration	20 October 2026 – 28 October 2026
Tableau Dashboard	https://public.tableau.com/app/profile/vijeta.nagda/viz/F1_Race_Intelligence/F1RaceIntelligenceDashboard
Live Web App	https://f1-race-intelligence-jbil.onrender.com/
GitHub	https://github.com/vijetanagda18/f1-race-intelligence

Table of Contents

1. INTRODUCTION 3
1.1 Project Overview 3
1.2 Purpose 3
2. IDEATION PHASE 4
2.1 Problem Statement 4
2.2 Empathy Map Canvas 4
2.3 Brainstorming 4
3. REQUIREMENT ANALYSIS 6
3.1 Customer Journey Map 6
3.2 Solution Requirement 6
3.3 Data Flow Diagram 6
3.4 Technology Stack 7
4. PROJECT DESIGN 8
4.1 Problem Solution Fit 8
4.2 Proposed Solution 8
4.3 Solution Architecture 8
5. PROJECT PLANNING & SCHEDULING 9
5.1 Project Planning 9
6. FUNCTIONAL AND PERFORMANCE TESTING 10
6.1 Performance Testing 10
7. RESULTS 11
7.1 Output Screenshots 11
7.2 Key Findings 14
8. ADVANTAGES & DISADVANTAGES 15
9. CONCLUSION 16
10. FUTURE SCOPE 17
11. APPENDIX 18
Source Code 18
Dataset 18
GitHub & Project Demo Links 18

1. INTRODUCTION

1.1 Project Overview

Formula 1 (F1) produces a large amount of data every season, including race results, qualifying times, lap and sector times, tyre strategies, pit stops and championship standings. In raw table form this data is hard to interpret. The F1 Race Intelligence Dashboard turns it into an interactive visual analytics product built with Tableau Public and delivered through a Flask web application.

The dashboard summarises four headline KPIs: 10 competing teams, 17 seasons in play (2010 to 2026), 6,464 race outcomes logged and 24 circuits showcased. Around these KPIs it presents nine visualisations: a global driver distribution map, driver reliability analysis, driver performance leaderboard, points trajectory, team performance ranking, team championship analysis, country-wise race participation, tyre compound distribution and podium achievements by driver and team.

Tableau Public dashboard:

https://public.tableau.com/app/profile/vijeta.nagda/viz/F1_Race_Intelligence/F1RaceIntelligenceDashboard

Live Flask web application:

https://f1-race-intelligence-jbil.onrender.com/

1.2 Purpose

The purpose of this project is to build a data-driven dashboard that makes F1 race data easy to understand and act on. It aims to:

- Present driver and constructor performance clearly and visually.
- Let users explore data by season, driver, team, DNF reason and circuit.
- Reveal patterns such as dominant teams, reliability problems, tyre usage and race distribution across countries.
- Demonstrate practical skills in data preparation, visual design, dashboard interactivity and web deployment.

2. IDEATION PHASE

2.1 Problem Statement

F1 data is rich but scattered across many tables. Fans and analysts often depend on static results pages that do not show trends, comparisons or context. They cannot easily answer questions such as: Which team scored the most points overall? Which drivers suffer the most retirements, and why? How are tyre compounds used across races? A single interactive tool is needed to answer such questions quickly.

Problem Statement Element	Description
I am	An F1 fan, student or data analyst
I am trying to	Understand race, driver and team performance quickly
But	The data is scattered, static and hard to compare
Because	There is no single interactive visual summary
Which makes me feel	Overloaded and unsure of what the data really shows

2.2 Empathy Map Canvas

The empathy map captures the needs of the primary user: an F1 enthusiast or analyst using the dashboard.

Quadrant	Insights
Says	"Which team is the strongest?" "Why do some drivers retire so often?" "Which tyres are used most?"
Thinks	Wants quick, trustworthy answers without opening many tables.
Does	Checks results, compares drivers and teams, discusses findings with others.
Feels	Excited by competition; frustrated by cluttered or static reports.
Pains	Scattered data, no filters, difficulty comparing across seasons.
Gains	Clear visuals, interactive filters, fast comparison and shareable insights.

2.3 Brainstorming

Ideas were grouped by theme and prioritised by value and feasibility. The high-priority ideas were implemented in the final dashboard.

Theme	Ideas	Priority / Status
Headline KPIs	Teams, seasons, race outcomes, circuits	High – implemented
Driver analysis	Leaderboard, reliability (DNFs by reason), podiums	High – implemented
Team analysis	Team ranking, championship points	High – implemented

Theme	Ideas	Priority / Status
Race and circuit insights	Country-wise participation, driver distribution map	High – implemented
Strategy insights	Tyre compound distribution	Medium – implemented
Trends	Points trajectory by season	High – implemented
Interactivity	Season (year) filter, DNF reason legend filter, tooltips	High – implemented
Web delivery	Flask web app hosted on Render	Medium – implemented
Advanced	Predictive models, lap-by-lap analysis	Future scope

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Awareness	Exploration	Analysis	Action
User activity	Receives the dashboard or web app link	Opens dashboard and reads the KPIs	Filters by year, DNF reason, team or driver	Draws conclusions and shares the link
Touchpoint	Flask web app / Tableau Public	Overview with KPI cards	Charts and tooltips	Share / revisit
Emotion	Curious	Engaged	Satisfied	Confident
Opportunity	Clear title and short description	Simple, uncluttered layout	Fast, responsive filtering	Easy sharing by link

3.2 Solution Requirement

Functional requirements

- Display headline KPIs: competing teams, seasons, race outcomes and circuits.
- Show driver and team rankings through charts.
- Provide a year filter and a DNF reason filter, plus tooltips on charts.
- Show geographic distribution of races on a map.
- Publish the dashboard publicly and make it reachable through a Flask web application.

Non-functional requirements

- Usability: clean layout with clear chart titles.
- Performance: fast loading and quick response to filter changes.
- Accuracy: figures must match the source dataset.
- Availability: hosted on Tableau Public and Render, accessible from any modern browser.
- Maintainability: source code and documentation stored in a GitHub repository.

3.3 Data Flow Diagram

The project uses an Excel dataset (F1_Racing_Project_Dataset.xlsx) with five sheets:

Sheet	Rows	Description
race_results	6,464	42 columns: race, circuit, weather, grid and finish position, points, DNF and reason, pit stops, tyre strategy, laps led
championship_standings	311	Season-wise driver standings: points, wins, podiums, pole positions, DNFs, champion flag
qualifying_results	3,130	Q1, Q2 and Q3 times, gap to pole, sector times, top speed
lap_intelligence	3,814	Lap and sector times, tyre compound and age, DRS, speed trap, pit lap flag

Sheet	Rows	Description
circuits	29	Circuit name, country, city, laps, lap length, type, altitude, direction

Data flow:

1. Source	2. Ingestion	3. Preparation	4. Visualisation	5. Delivery
Excel dataset with five sheets	Imported into Tableau as a data source	Cleaning, joins and calculated fields (e.g., DNF count, points)	Worksheets combined into one dashboard with filters	Published on Tableau Public; Flask app deployed on Render

When a user applies a filter or clicks a mark, Tableau re-queries the data and updates all linked views.

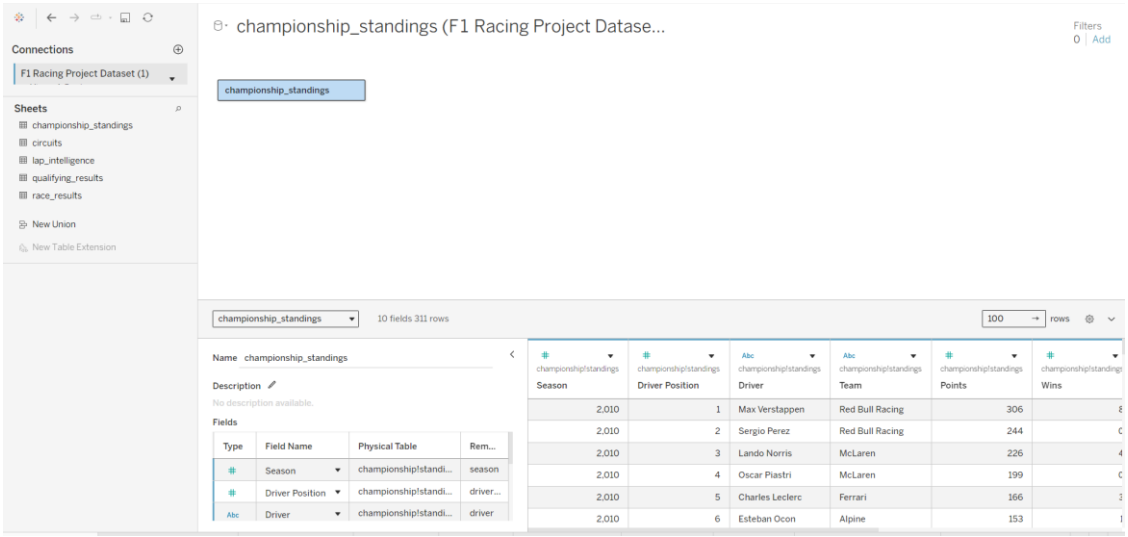


Figure 3.1: Tableau Data Source tab showing the five imported sheets

3.4 Technology Stack

Component	Technology / Details
Visualisation	Tableau Public (bar, stacked bar, line, treemap, pie, bubble and map charts)
Map layer	Mapbox / OpenStreetMap base map in Tableau
Data source	Microsoft Excel workbook (.xlsx) with five sheets
Data preparation	Joins, data cleaning and calculated fields in Tableau
Web application	Python Flask
Hosting	Tableau Public Cloud (dashboard) and Render (Flask app)
Version control	Git and GitHub
Access	Any modern web browser

4. PROJECT DESIGN

4.1 Problem Solution Fit

Aspect	Description
Customer segment	F1 fans, students and sports data analysts
Problem	Scattered, static race data that is hard to compare
Existing alternatives	Results websites, spreadsheets and Wikipedia tables
Our solution	One interactive dashboard with linked charts, filters and tooltips
Why it fits	Gives instant visual answers to common F1 performance questions
Channel	Public Tableau link and Flask web application

4.2 Proposed Solution

The proposed solution is an interactive Tableau dashboard that consolidates the F1 dataset into KPI cards, rankings, trends and comparisons, delivered through a Flask web application. Key features are:

- Four headline KPIs: 10 teams, 17 seasons, 6,464 race outcomes and 24 circuits.
- Global driver distribution map showing race counts per country.
- Driver reliability analysis showing DNFs stacked by reason, with a year filter.
- Driver leaderboard, points trajectory and team ranking charts.
- Country-wise race participation treemap and tyre compound distribution pie.
- Podium achievements by driver and team shown as a packed bubble chart.

4.3 Solution Architecture

Layer	Description
Data layer	Excel workbook with race results, standings, qualifying, lap and circuit data.
Processing layer	Data cleaning, joins and calculated fields prepared in Tableau.
Presentation layer	Worksheets combined into the F1 Race Intelligence Dashboard on Tableau Public.
Application layer	Flask web application with Dashboard and Story pages that embeds the Tableau dashboard and presents filters (DNF reason, race year, Top N drivers) to users.
Deployment layer	Tableau Public Cloud and Render hosting; source code on GitHub.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was carried out from 20 October 2026 to 28 October 2026.

No.	Phase	Activities	Dates	Status
1	Requirement gathering	Define objectives, users and questions to answer	20 Oct	Done
2	Data collection	Obtain and review the F1 dataset (five sheets)	21 Oct	Done
3	Data preparation	Clean data, join tables, create calculated fields	22 Oct	Done
4	Visualisation	Build worksheets and charts in Tableau	23–24 Oct	Done
5	Dashboard design	Arrange layout, filters, tooltips and formatting	25 Oct	Done
6	Web app and deployment	Build Flask app, push to GitHub, deploy on Render	26 Oct	Done
7	Testing	Verify accuracy, interactivity and performance	27 Oct	Done
8	Documentation	Prepare final project report	28 Oct	Done

Team ID: EN25CS3011046 | Developer: Vijeta Nagda

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The dashboard and web application were tested for functionality, data accuracy and responsiveness.

ID	Test Case	Expected Result	Result	Status
T1	Tableau dashboard loads from link	Dashboard opens without errors	As expected	Pass
T2	Flask app loads from live link	Web app opens on Render	As expected	Pass
T3	Year filter (2010–2026)	All linked charts update	As expected	Pass
T4	DNF reason legend filter	Reliability chart highlights selected reason	As expected	Pass
T5	Tooltips	Map and charts show details on hover	As expected	Pass
T6	KPI accuracy	Teams, seasons, races and circuits match source data	As expected	Pass
T7	Browser compatibility	Works on Chrome, Edge and Firefox	As expected	Pass

Performance observations: the dataset is moderate in size (about 6,500 race rows), so charts render quickly and filters respond without noticeable delay. On the free Render plan the Flask app may take a few seconds to wake up after a period of inactivity.

7. RESULTS

7.1 Output Screenshots

The screenshots below show the final dashboard and its main visualisations, with the key findings from each.

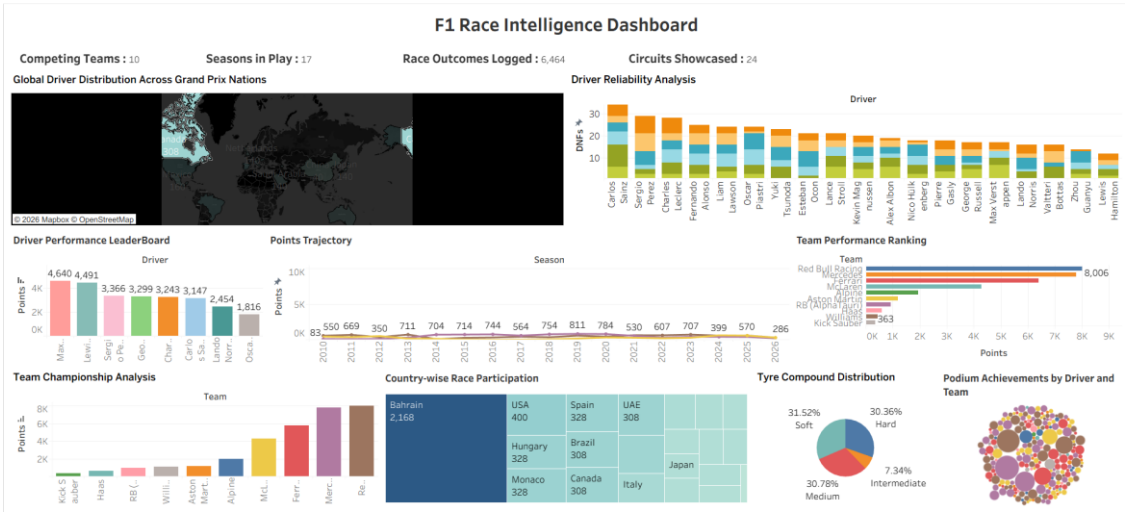


Figure 7.1: F1 Race Intelligence Dashboard – complete view

The dashboard opens with four KPIs (10 teams, 17 seasons, 6,464 race outcomes, 24 circuits) followed by nine linked visualisations.

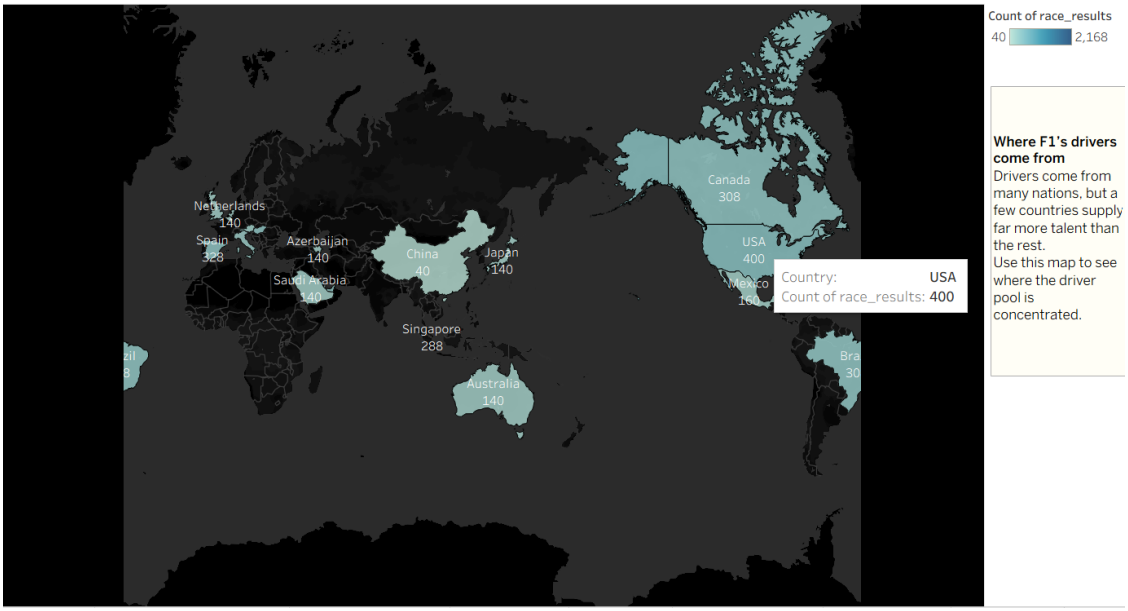


Figure 7.2: Global driver distribution across Grand Prix nations

The map shows race counts per country, using colour intensity. Bahrain has the highest count (2,168), followed by the USA (400); Spain, Hungary and Monaco have 328 each. Hovering over a country shows its exact count.

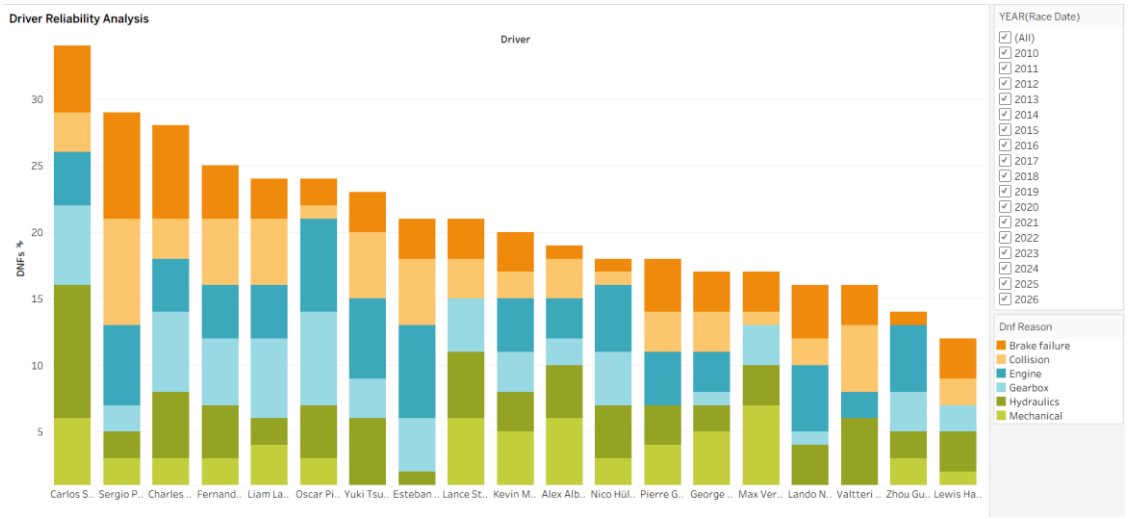


Figure 7.3: Driver reliability analysis (DNFs by reason)

This stacked bar chart ranks drivers by number of DNFs and splits each bar by cause: brake failure, collision, engine, gearbox, hydraulics and mechanical. Carlos Sainz has the most retirements (about 34), followed by Sergio Perez and Charles Leclerc. A year filter lets the user focus on specific seasons.

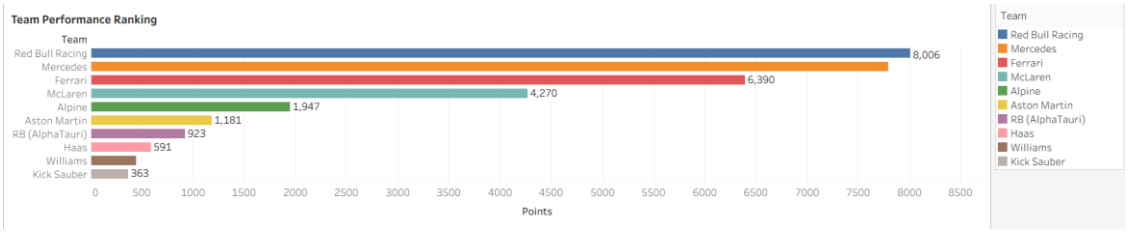


Figure 7.4: Team performance ranking

Red Bull Racing leads with 8,006 points, followed by Mercedes and Ferrari (6,390). McLaren has 4,270 and Alpine 1,947, while Kick Sauber has the lowest total at 363.

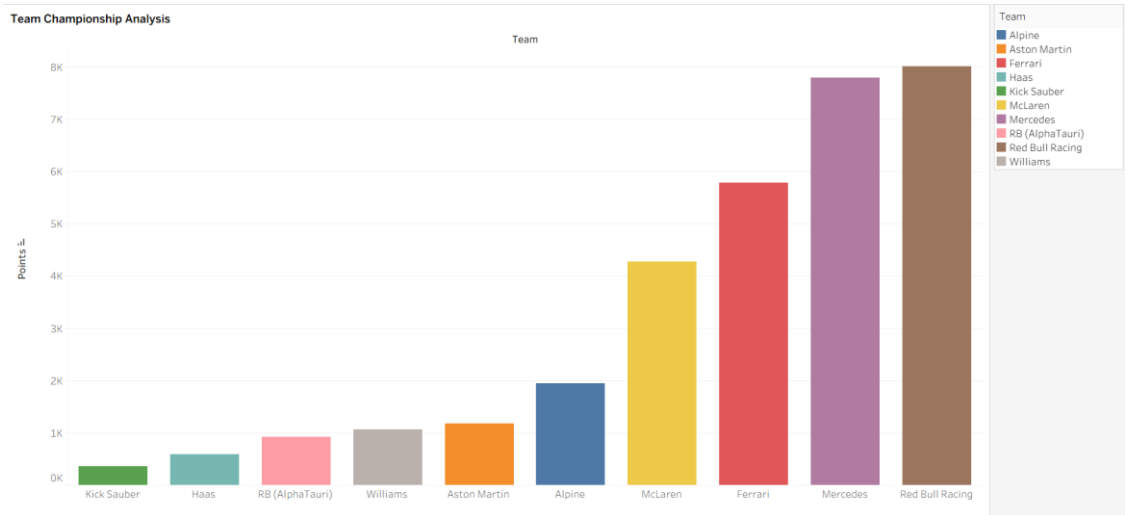


Figure 7.5: Team championship analysis

The column chart compares total championship points per team. Red Bull Racing, Mercedes and Ferrari form the leading group, well ahead of the midfield teams.

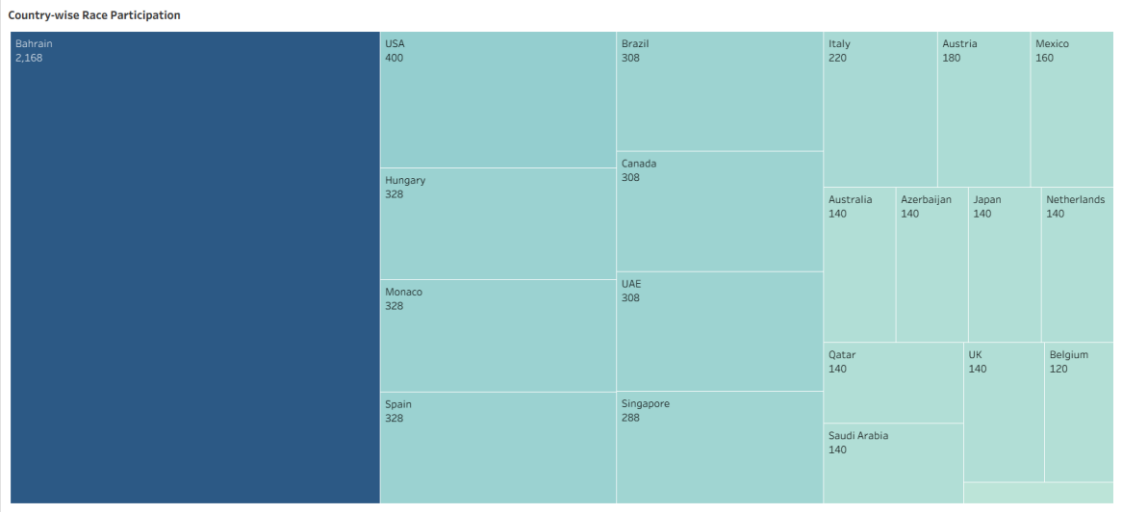


Figure 7.6: Country-wise race participation

The treemap shows Bahrain as by far the largest block (2,168 race entries), with the USA (400), Brazil, Canada and UAE (308 each) and several other countries below.

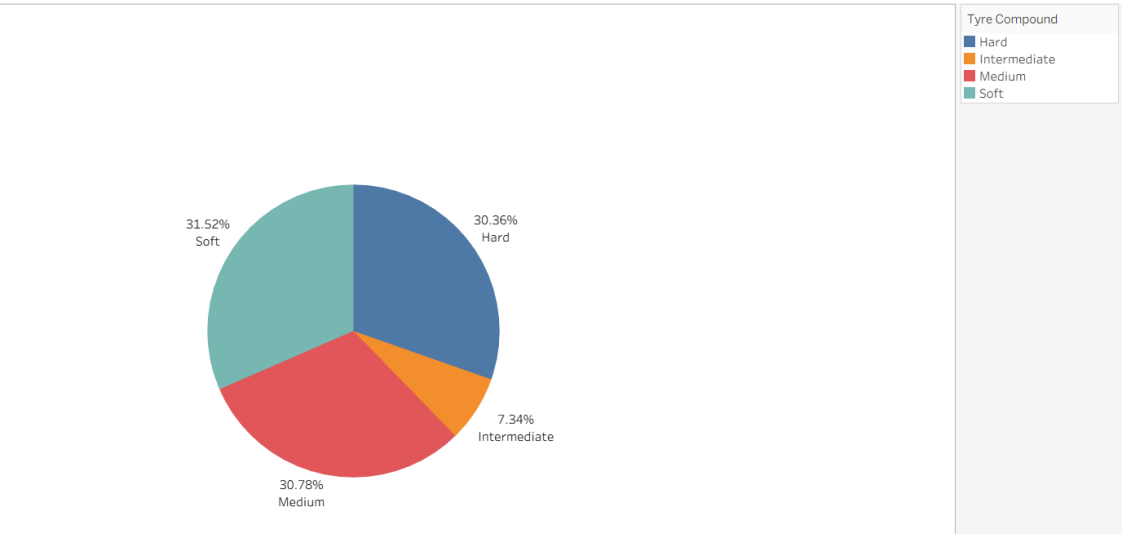


Figure 7.7: Tyre compound distribution

Tyre usage is evenly split: Soft 31.52%, Medium 30.78% and Hard 30.36%, while Intermediate tyres are used only 7.34% of the time.

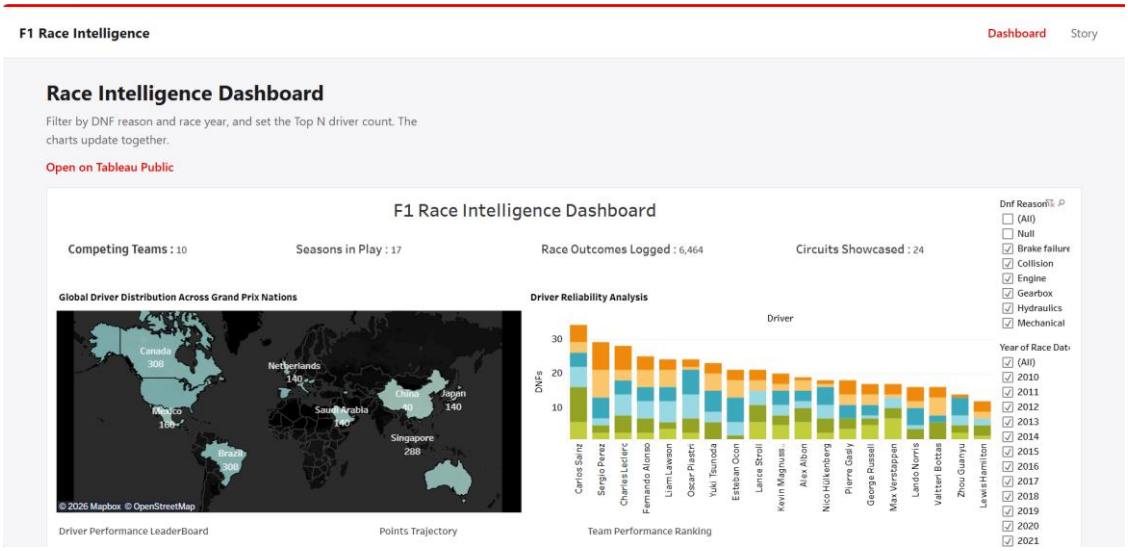


Figure 7.8: Flask web application with embedded Tableau dashboard

The Flask web application, hosted on Render, presents the dashboard under the title "F1 Race Intelligence". It has two pages, Dashboard and Story, and an "Open on Tableau Public" link. Users can filter by DNF reason and race year, and set the Top N driver count, and the charts update together.

7.2 Key Findings

- Red Bull Racing is the highest-scoring team (8,006 points), ahead of Mercedes and Ferrari.
- Max Verstappen and Lewis Hamilton lead the driver leaderboard with 4,640 and 4,491 points respectively.
- Engine, hydraulics and mechanical issues are the most common retirement causes in the dataset.
- Soft, medium and hard tyres are used in nearly equal proportions.
- Bahrain accounts for the largest share of race entries.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive and easy to explore without technical knowledge.
- Combines multiple data sheets into one clear view.
- Quick comparison of drivers, teams, tyres and countries.
- Free, publicly accessible hosting through Tableau Public and Render.
- Source code available on GitHub for reuse and improvement.

Disadvantages

- Tableau Public workbooks are publicly visible, so they are unsuitable for confidential data.
- Data is static and must be updated manually for new races.
- The free Render plan may cause slow start-up after inactivity.
- The dashboard does not include predictive analytics.

9. CONCLUSION

The F1 Race Intelligence Dashboard converts complex Formula 1 data into a clear, interactive and insightful tool. It meets its objectives of simplifying data exploration, comparing drivers and teams, analysing reliability and tyre usage, and presenting race distribution across countries. By combining Tableau visualisations with a Flask web application hosted on Render, the project shows an end-to-end workflow from raw data to a publicly accessible analytics product, and it strengthened practical skills in data preparation, dashboard design and deployment.

10. FUTURE SCOPE

- Connect live or automatically refreshed data using APIs such as FastF1 or Ergast.
- Add lap-by-lap analysis, pit-stop strategy and sector comparisons using the lap and qualifying data.
- Add predictive models for race and championship outcomes.
- Provide mobile-optimised dashboard layouts.
- Add head-to-head driver comparison and circuit-level drill-downs.

11. APPENDIX

Source Code

The Flask application source code is available in the GitHub repository below. The Tableau dashboard was built using Tableau's visual interface, and its calculated fields are stored inside the Tableau workbook.

Dataset

F1_Racing_Project_Dataset.xlsx – an Excel workbook with five sheets: race_results (6,464 rows), championship_standings (311 rows), qualifying_results (3,130 rows), lap_intelligence (3,814 rows) and circuits (29 rows). The dataset file is also included in the GitHub repository / project submission.

GitHub & Project Demo Links

Item	Link
Tableau Public Dashboard	https://public.tableau.com/app/profile/vijeta.nagda/viz/F1_Race_Intelligence/F1RaceIntelligenceDashboard
Live Flask Web App	https://f1-race-intelligence-jbil.onrender.com/
GitHub Repository	https://github.com/vijetanagda18/f1-race-intelligence
Team ID	EN25CS3011046
Developer	Vijeta Nagda